03 21826 Autumn 2025
LH The Life and Death of Stars

Lecture 20

(JCD 14)

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Previous lecture

- Evolutionary tracks: curve in HR diagram for a fixed mass and chemical composition and with varying age
- Isochrones: curve in HR diagram for a fixed age and chemical composition but for varying mass
- Stellar clusters lie along an isochrone and observations of clusters can thus measure precise ages

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Outline

Massive star evolution

Supernovae
- Core-collapse
- Shock wave
- Observing supernovae

Chemical enrichment

Supermassive star evolution

Summary

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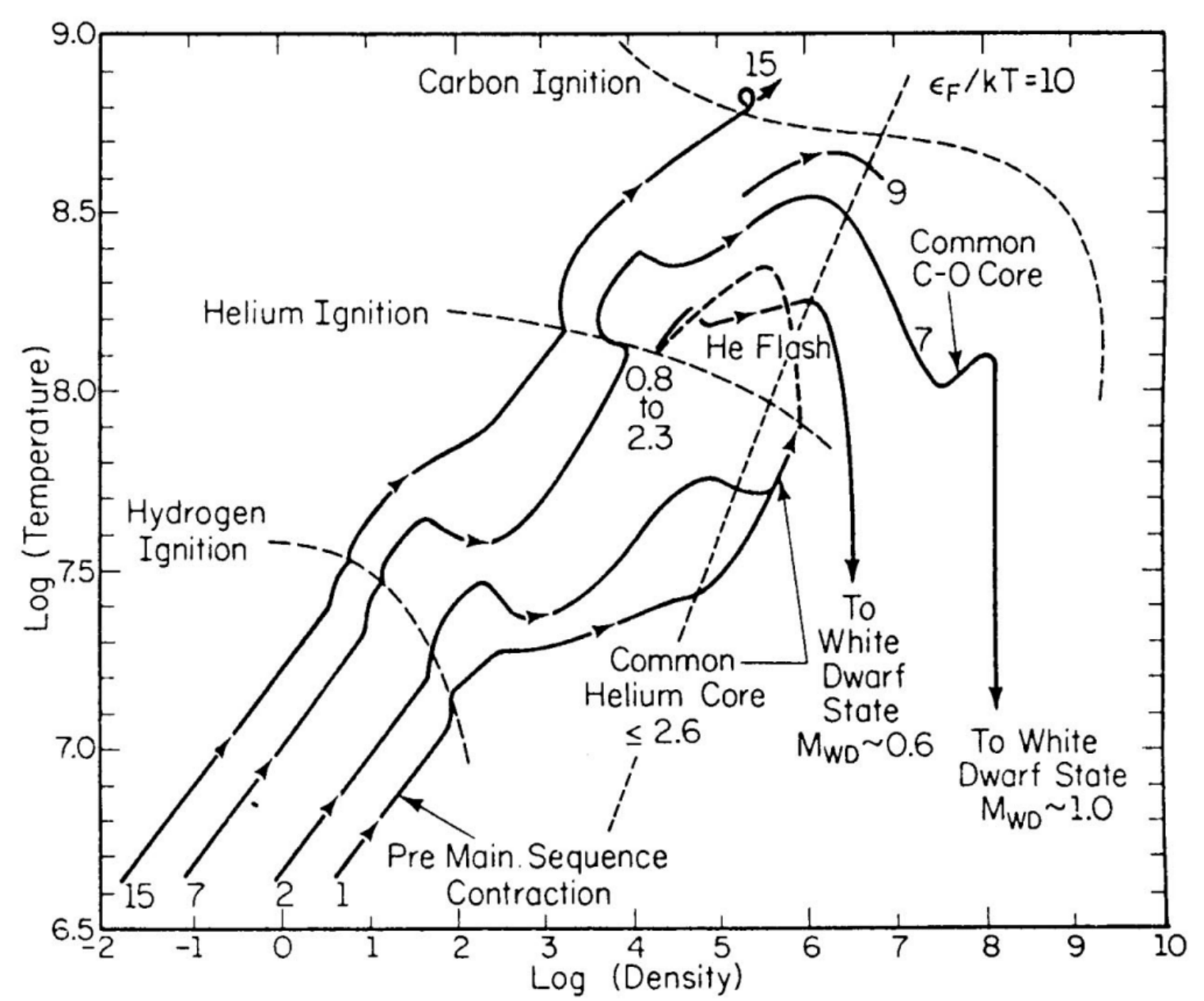
Stellar evolution - degenerate cores



Credit: JCD - Fig 12.7

Massive star evolution ($M_* > 10\,\mathrm{M}_\odot$)

- Initial evolution is the same as moderate mass stars, till there is an inert C-O core with He and H shells burning.
- C-O core then does not become degenerate before C fusion temperature is reached
- Evolution proceeds in same way as before, with core nuclear fusion alternating with inert core that gravitationally contracts and heats
- Result is an onion-like structure of multiple shells of fusion and shells of vastly different chemical composition

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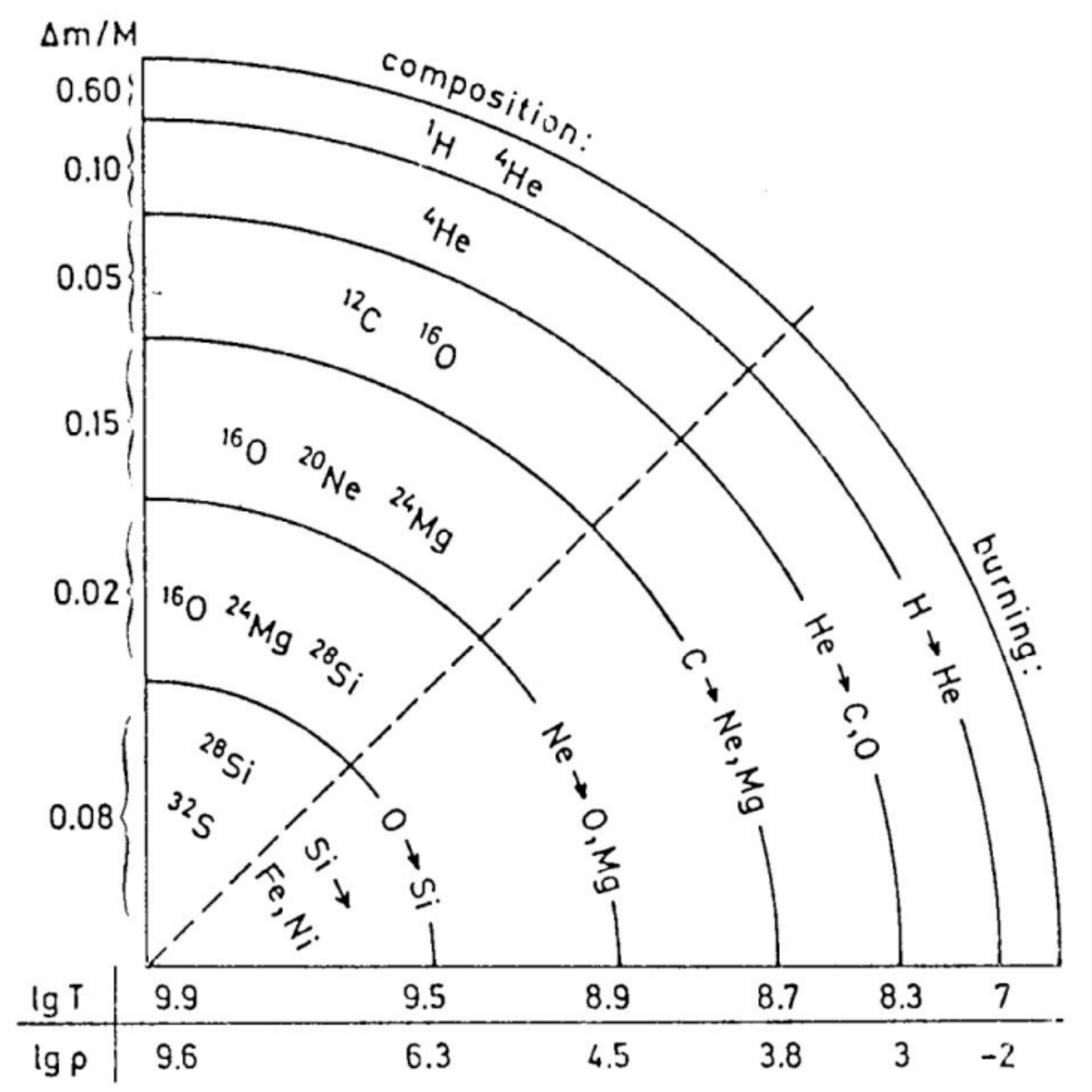
Onion-like structure



Late stages nuclear fusion

- Which elements will fuse depends on the nuclear binding energy - Remember lecture 11
- Order of fusion is carbon, neon, oxygen, "silicon" - Remember lecture 13
- The fusion stages timescales grow shorter and shorter - Remember lecture 2
- The process stops when iron group elements are reached as fusing further elements requires energy (and is thus fission, not fusion)

"Silicon" burning

- After the exhaustion of oxygen in the core and the subsequent gravitational contraction, temperatures are so high that photo-dissociation becomes dominant
- This is the process where high-energetic photons (due to high thermal energy from the high temperatures) break up nuclei and release ${}^{4}_{2}\mathrm{He}$
- Note this is also the reason for neon fusion happening prior to oxygen fusion
- The released ${}^{4}_{2}\mathrm{He}$ can then be captured by other elements (or the same one) and further release energy
- Result is complex network of reactions, all in both directions, and mistakenly called "silicon burning"

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Core contraction

- After all core fusion stops: Iron core is inert so, as always, it contracts
- This means temperature (virial theorem) and density keep rising
- Core becomes electron-degenerate, where most mass (density) is given by the nuclei but most pressure comes from the electron degeneracy pressure
- With temperatures rising, the electrons also become relativistic
- It can be shown, by balancing the pressures of HE and relativistic electrons, that a mass limit must exist beyond which the electron pressure can not balance the core
- This is Chandrasekhar limit - See lecture 21 - and is about $1.4\,M_\odot$ at this stage of "silicon" burning

Core-collapse

- The virial theorem requires that gravitational potential energy from contraction is converted into internal kinetic energy.
- This can be a stable and self-correcting process where the increase in temperature leads to an exothermic nuclear reaction with a further increase in internal energy and an increase in pressure opposing the contraction.
- It can also be an unstable process where it triggers an energy absorbing process, kinetic energy is absorbed and the effectiveness of the pressure decreases. Under these circumstances, the contraction can turn into a collapse.

Energy-absorbing processes

- There are two possible energy-absorbing processes.
- One is the photodisintegration of atomic nuclei where kinetic energy unbinds atomic nuclei.
- Second is electron capture where the kinetic energy of the degenerate electrons is converted into kinetic energy of electron neutrinos which escape from the core.
- Both processes happen in the degenerate iron core of an evolved massive star.

Core-collapse timescale

- Collapse goes into almost free fall as it is unopposed
- Happens on a dynamical timescale (lecture 1)

$$t_{ff} \approx \sqrt{\frac{1}{G\rho}}$$

- This is on the order of milliseconds for the dense iron core

Photodisintegration

- Photons have enough energy to break up iron

$$^{56}\mathrm{Fe} \rightarrow 13\,^{4}\mathrm{He} + 4\,n$$

- Helium also breaks up at even higher temperatures

$$^{4}\mathrm{He} \rightarrow 2\,p + 2\,n$$

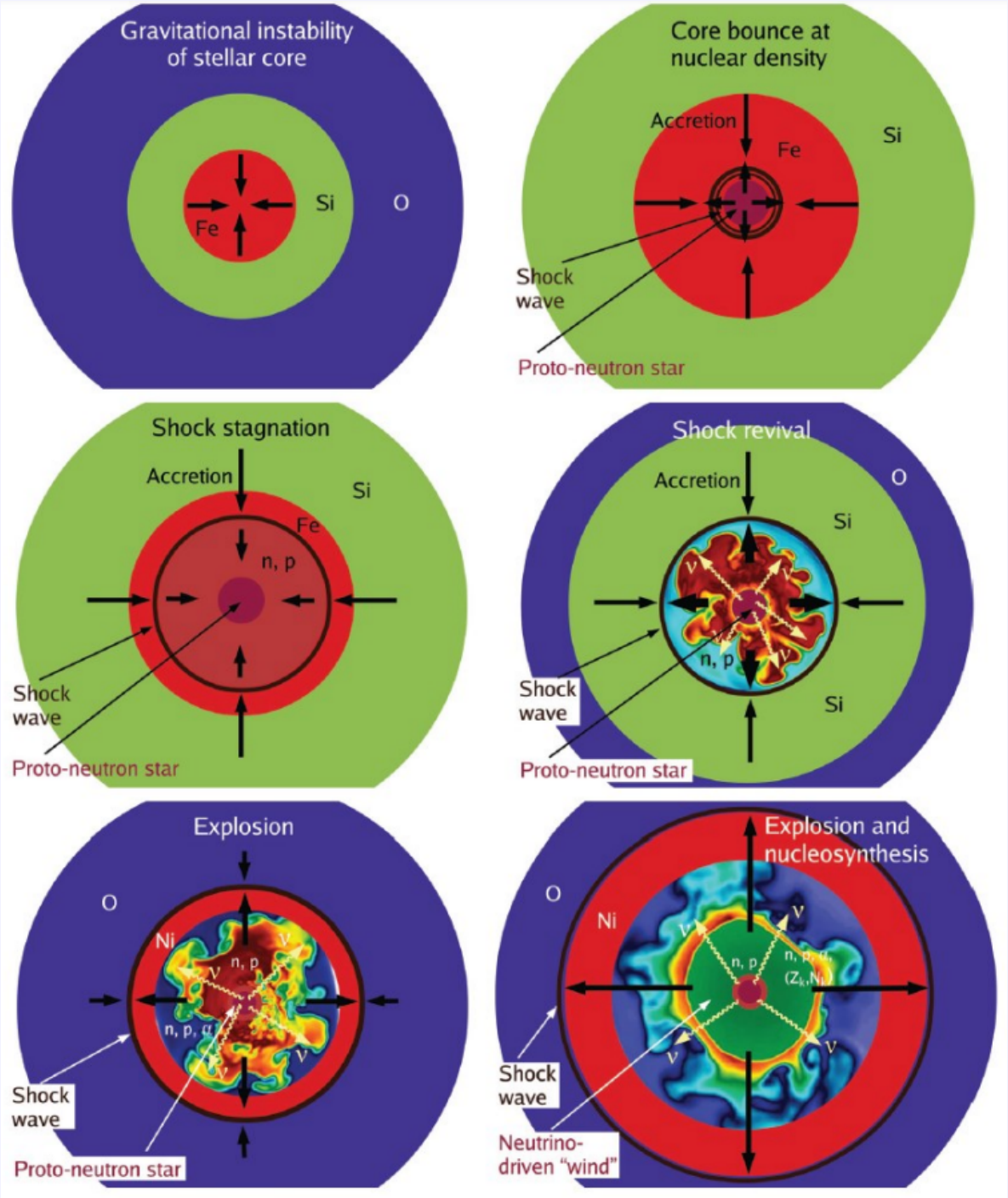
- Both are endothermic reactions
- Core becomes a "soup" of neutrons

Collapse stops

- Density keeps rising
- Core is now mostly made of neutrons - proto-neutron star
- Neutrino's become trapped in the core
- Finite compressability of nuclear matter (neutrons) will bring the collapse to a halt

Shock wave

- At the sudden stop to core collapse, a shock wave is created at the core surface
- The shock wave moves outwards
- The matter through which the shock wave moves will be dissociated due to the high temperatures
- Recent numerical simulations show that the shock wave first stagnates but is then revived likely due to neutrino interactions
- This then causes the star's outer regions to explode = Core-Collapse Supernova
- A neutron star is left as a remnant (lecture 21)
- Note that the whole process happens on a timescale of seconds

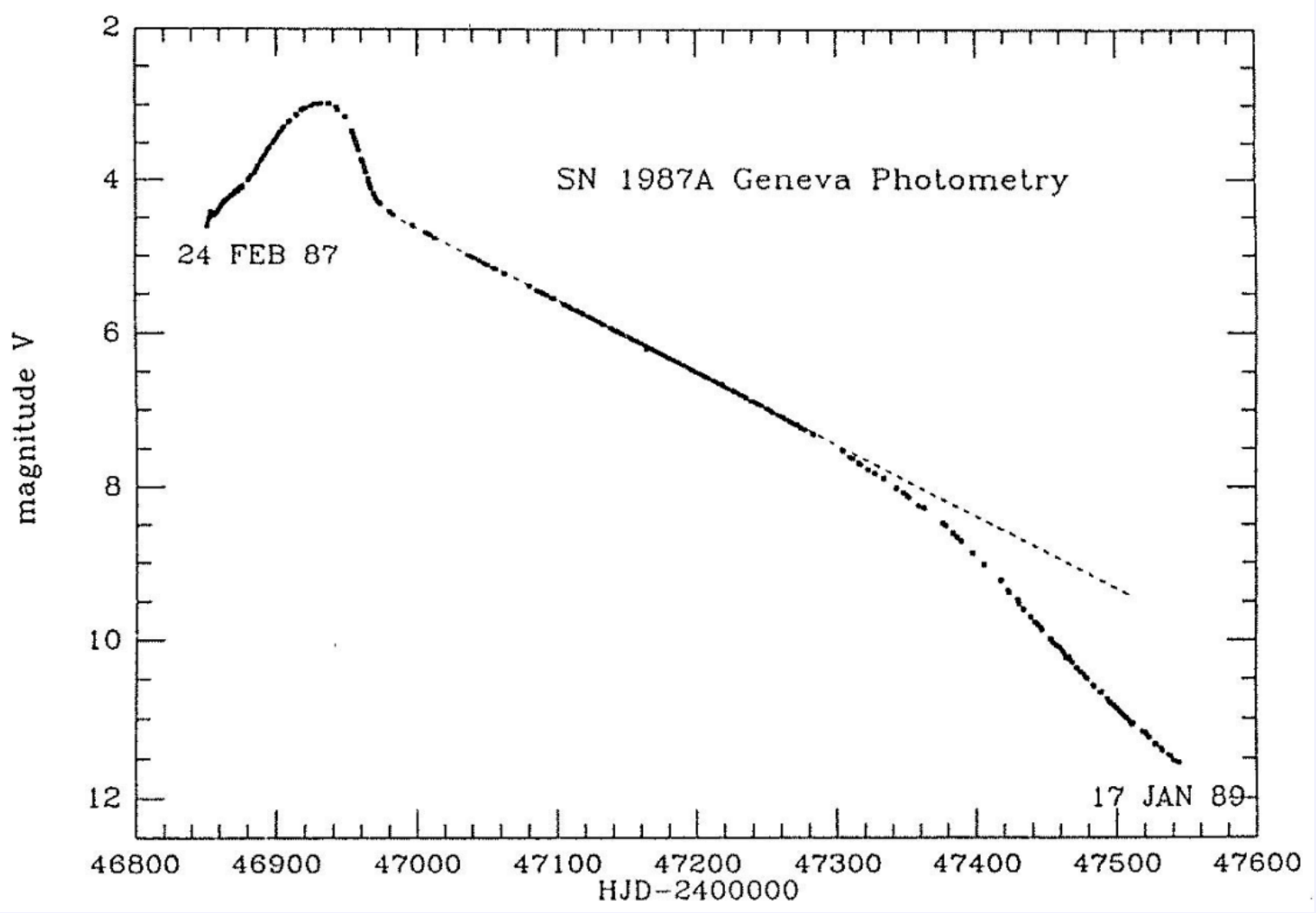

Gravitational instability of stellar core
Fe
Si
O
Core bounce at nuclear density
Accretion
Fe
Si
Shock wave
Proto-neutron star
Shock stagnation
Accretion
Si
Fe
n, p
Shock wave
Proto-neutron star
Shock revival
Accretion
O
Si
ν
n, p
Shock wave
Si
Proto-neutron star
Explosion
O
Ni
ν
n, p
n, p, α
Shock wave
Proto-neutron star
Explosion and nucleosynthesis
O
Ni
ν
n, p
Shock wave
Neutrino-driven "wind"

Credit: JCD

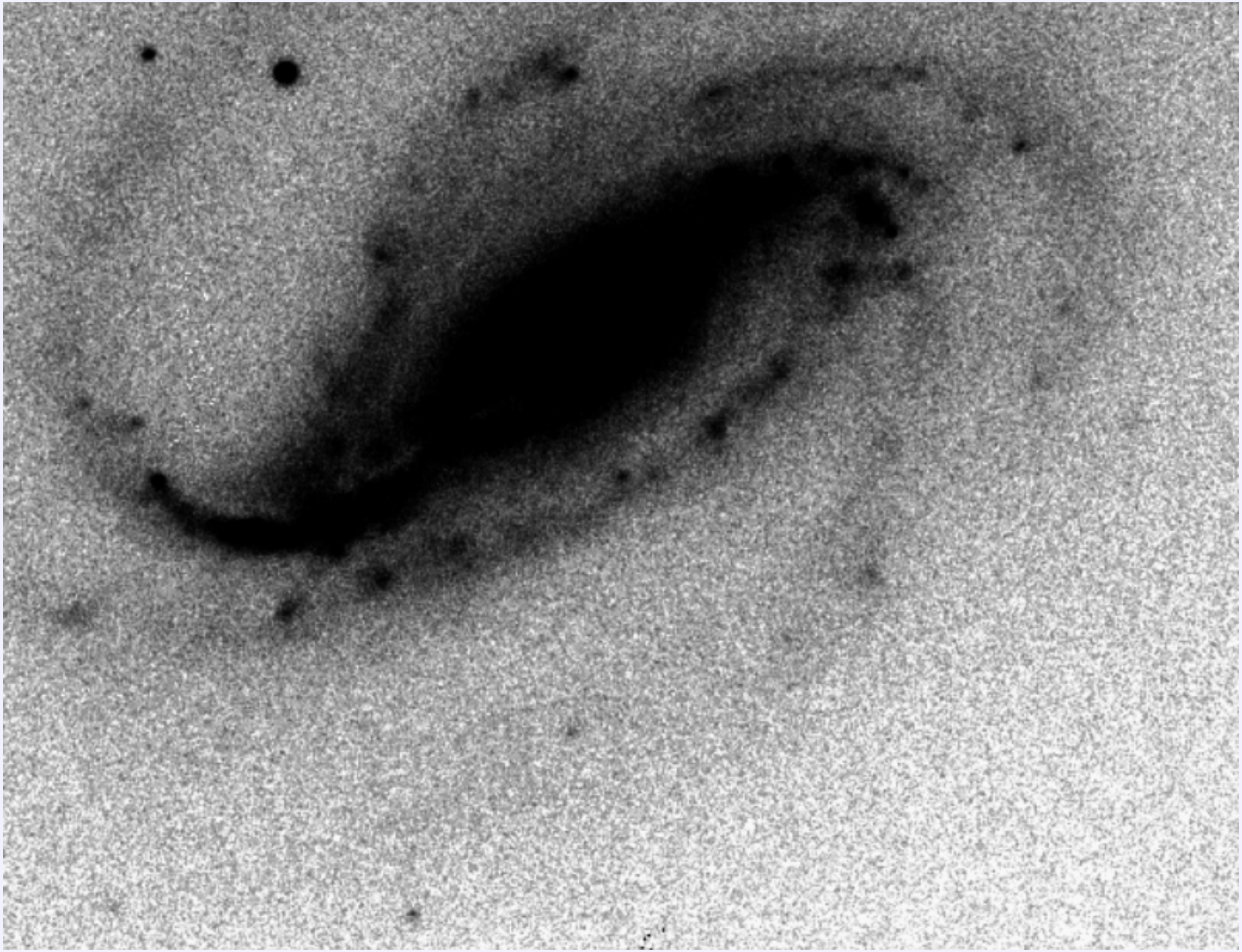
Observing supernovae

- Luminosity temporarily reaches $10^9 - 10^{10}\,L_\odot$ (equivalent to a galaxy)
- Remains bright for several months and then gradually dims
- Depending on what caused the explosion and the chemical composition, observed spectra of supernovae look different
- Easy to observe the aftermath, but rare to catch one during explosion
- Note that most observed supernovae are in other galaxies, not the Milky Way
- Even rarer to catch the onset: Recently the birth of a supernovae was for the first time observed accidentally by an amateur-astronomer.

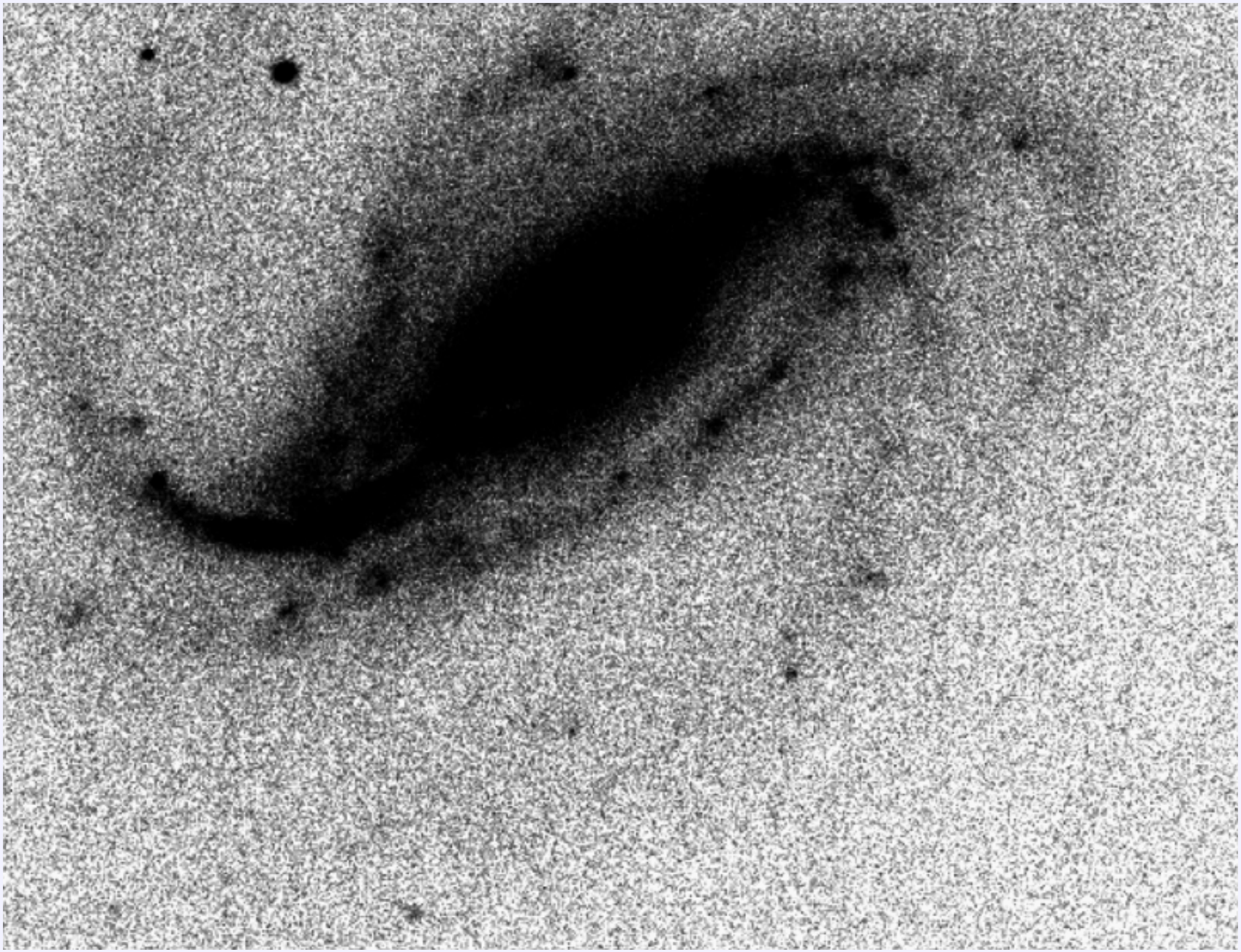
Observed brightness for SN1987A



Credit: JCD - Fig 14.6



(credit: Victor Buso - Nature)



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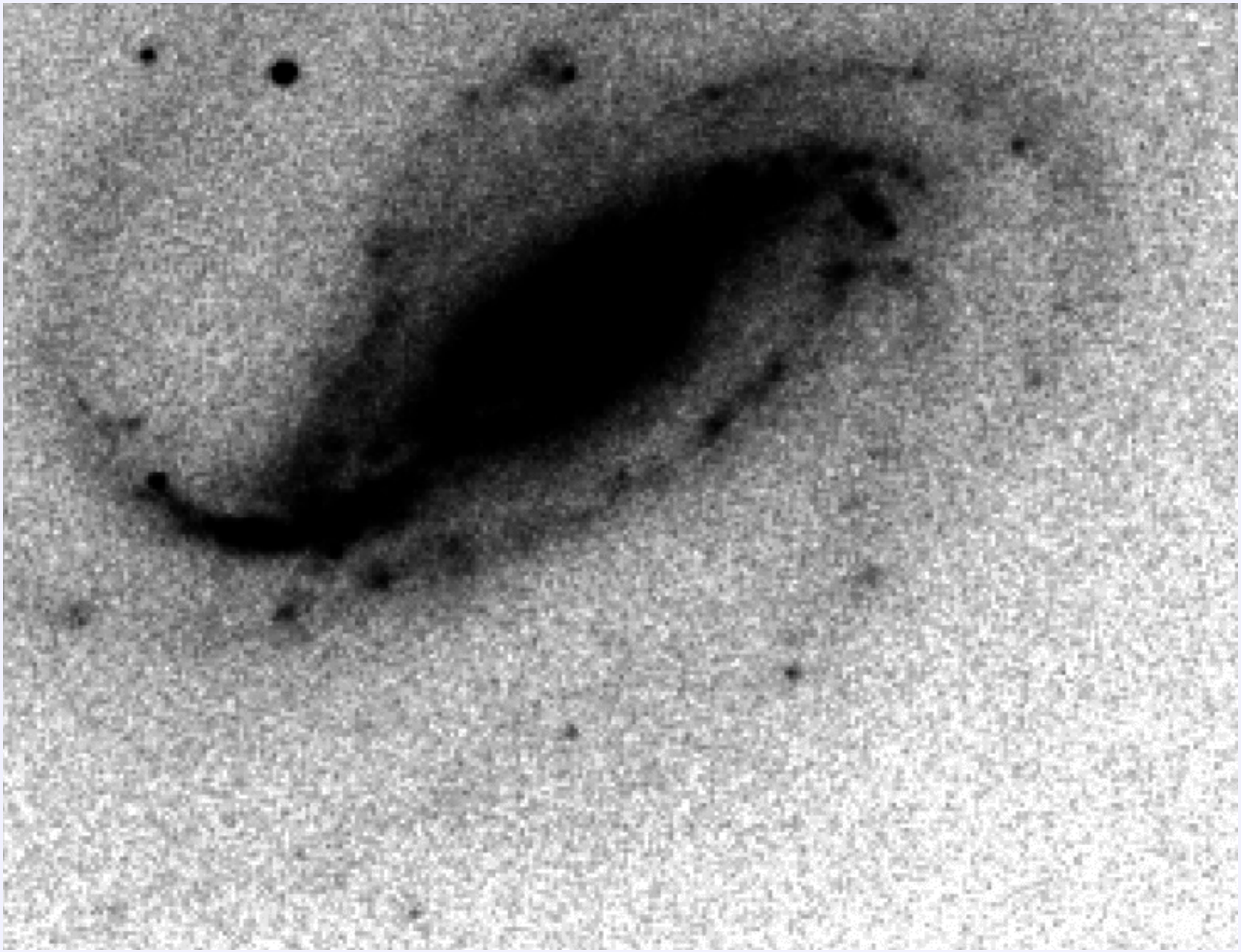

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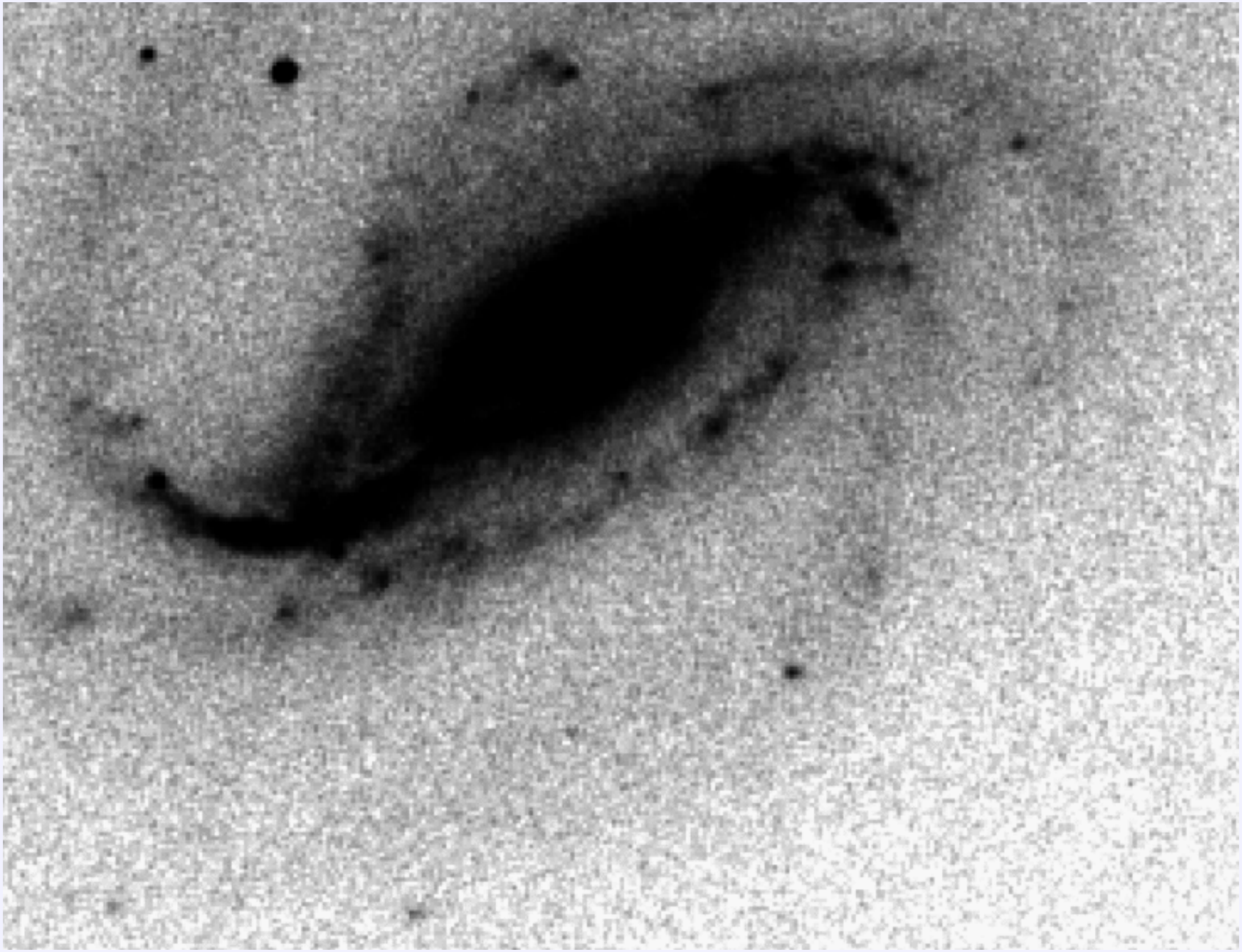

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Chemical enrichment of interstellar matter

- During all mass-loss events, ISM is enriched with heavy elements
- Mass-loss events are not just supernovae
- Mass-loss also happens at the planetary nebulae stage, or even by stellar winds during almost all stages of stellar evolution
- This chemical enrichment is the reason why younger stars are more metal-rich than older stars

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Supermassive star evolution ($M_* > 20\,\mathrm{M}_\odot$)

- For the same reasons there is a maximum mass for electron-degeneracy cores, there is a limiting mass for neutron-degeneracy cores (Chandrasekhar mass - see Lecture 21)
- If neutron star exceeds $\mathrm{M}_{\mathrm{Ch,n}}$, the collapse can't be stopped by the degeneracy pressure of the neutrons.
- Thus there will be complete collapse to a singularity $\Rightarrow$ Black hole

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Summary

- Massive star evolution creates onion-like structure in star
- Iron core will collapse in free fall after exceeding the Chandrasekhar mass
- Neutrons form in the core and rising density will stop the collapse
- A shock wave travels through the star and makes it explode, leaving a neutron star
- Supermassive stars will end with a black hole

NEXT LECTURE: White dwarfs and neutron stars